MAXWELL'S PROBLEM FOR THREE POINT CHARGES

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Abstract. We study the critical points of the potential $V(x) = \sum_{i=1}^{3} q_i|x - p_i|^{-2\alpha}$ of three point charges.

For charges of one sign we prove that V has at most four critical points, for every $\alpha > 0$ and in every dimension $n \geq 2$. The bound is sharp. The best bound known before was six, due to Gabrielov, Novikov, Novikov and Shapiro.

For charges of different signs we prove that the Coulomb potential in $\mathbb{R}^3$ has at most four critical points if the charges are not collinear, and at most four non-degenerate ones if they are. The same holds for every $\alpha \in [\frac{1}{4}, 2]$. Together, these results prove Maxwell's bound $(k-1)^2$ for $k = 3$ Coulomb charges: for all equilibria when the charges are not collinear, and for the non-degenerate ones when they are.

In the angles under which a critical point sees the sides of the triangle, the equilibrium equations separate. Whether a critical point is a minimum is decided by an explicit inequality, and two minima are excluded once the region defined by that inequality is convex. For charges of one sign this convexity follows from log-convexity. For charges of different signs it fails for large exponents, and we establish it by a rigorous interval-arithmetic computation.

1. Introduction

In §113 of his *Treatise* [7], Maxwell asserted that k point charges in $\mathbb{R}^3$ produce at most $(k-1)^2$ isolated equilibrium points. His argument is incomplete, and the question became known as Maxwell's problem [4].

Gabrielov, Novikov and Shapiro [4] proved the first finite bounds, using fewnomial theory. In particular, they showed that three positive charges have at most 12 non-degenerate equilibria, for every exponent $\alpha > 0$.

Very recently:

- Arathoon, Ball and Kvalheim [1] constructed five positive charges with at least $24 > 16$ non-degenerate equilibria. So Maxwell's bound fails in general.
- Gabrielov, Novikov, Novikov and Shapiro [5] improved 12 to 6 for three positive charges. They note that Maxwell's value 4 was known only for three charges of equal magnitudes [9].

See also [2, 3, 8, 10] for related work. We settle the three-charge case for charges of one sign and every exponent. For charges of different signs and the Coulomb potential we prove the same bound for all equilibria of non-collinear charges, and for the non-degenerate equilibria of collinear ones (Corollary 1.3).

Theorem 1.1 (charges of one sign)**.** *Let $\alpha > 0$, $n \geq 2$, $q_1, q_2, q_3 > 0$, and let $p_1, p_2, p_3 \in \mathbb{R}^n$ be distinct. Then the potential*

$$V(x) = \sum_{i=1}^{3} q_i \, |x - p_i|^{-2\alpha}$$

has at most four critical points in $\mathbb{R}^n \setminus \{p_1, p_2, p_3\}$.

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- *If the p_i are collinear, there are exactly two critical points.*
- *Otherwise, every critical point lies in the open triangle $T = p_1p_2p_3$. At most one of them is a local minimum of $V|_\Pi$ or degenerate, where Π is the plane of the charges. All the others are non-degenerate saddles of $V|_\Pi$, and there are at most three of them.*

The bound is attained, e.g. by three equal charges at the vertices of an equilateral triangle: the centre is a minimum in Π, and there are three saddles. Coulomb's law in $\mathbb{R}^3$ is the case $\alpha = 1/2$.

Theorem 1.2 (charges of different signs)**.** *Let q_1, q_2, q_3 be nonzero real numbers, not all of the same sign, placed at distinct points $p_1, p_2, p_3 \in \mathbb{R}^3$. Let $V(x) = \sum_i q_i|x - p_i|^{-1}$.*

(a) *If the p_i are not collinear, then V has at most four critical points, and all of them lie in the plane of the charges.*
(b) *If the p_i are collinear, then the critical points off the line form circles about it, and V has at most four non-degenerate critical points.*

Corollary 1.3 (Maxwell's bound for three charges)**.** *Three point charges in $\mathbb{R}^3$ with the Coulomb potential have at most four non-degenerate equilibrium points. If the charges are not collinear, they have at most four equilibrium points altogether.*

Proof. For charges of one sign this is Theorem 1.1 with $\alpha = \frac{1}{2}$, which counts all critical points. Otherwise it is Theorem 1.2. □

Theorem 1.4 (other exponents)**.** *Parts (a) and (b) of Theorem 1.2 also hold for $V = \sum_i q_i|x - p_i|^{-2\alpha}$ for every $\alpha \in [\frac{1}{4}, 2]$: part (a) in $\mathbb{R}^n$ for every $n \geq 2$, and part (b) in $\mathbb{R}^n$ for $n \geq 3$, where the critical points off the line form $(n-2)$-spheres about it.*

For $\alpha = 1, \frac{3}{2}, 2$ these are the Newtonian potentials of $\mathbb{R}^4, \mathbb{R}^5, \mathbb{R}^6$.

The proof of Theorem 1.1 does not use a computer. Theorems 1.2 and 1.4 rely on one computer-assisted step, Proposition 12.1, which is verified by interval arithmetic.

Idea of the proof. For a point x there are unique charges, up to a common factor, for which x is an equilibrium. The proof makes three observations about them.

(1) They take a separated form in the angles under which x sees the sides of the triangle; see (3). This rests on Euler's theorem on pedal triangles. It turns the equilibrium equations into the gradient equations of a function whose terms each depend on one angle.
(2) Whether x is a minimum is decided by an explicit inequality $E < (b+1)^2/b$ in these angles, with $b = 2\alpha + 1$ (Lemma 4.1).
(3) Where this inequality holds, the separated function is strictly convex (or concave). So two minima for the same charges are impossible once the region it defines is convex.

For charges of one sign, E is a sum of products of log-convex functions, so the region is convex (Lemma 4.2), and a Morse count gives at most three saddles.

For charges of different signs, the minima lie outside the triangle: beyond the side opposite the odd charge, outside the circumcircle, around the excentre. One product now enters E with a minus sign, and the region is not convex for large exponents (Remark 14.1). We describe its boundary in closed form through a strip model of the hyperbolic plane. We then reduce its convexity to the positivity of an explicit function of three variables (four, when the exponent varies), and verify that positivity by interval arithmetic (Section 12).

Part I (Sections 2–6) proves Theorem 1.1. Part II (Sections 7–14) proves Theorems 1.2 and 1.4.

Part I. Charges of one sign

2. Reduction

Throughout Part I, $\beta = 2\alpha + 2$, $b = \beta - 1 = 2\alpha + 1 > 1$, and

$$c = \frac{\beta - 1}{\beta^2} = \frac{b}{(b+1)^2}.$$

Write $e_2(v) = v_1v_2 + v_2v_3 + v_3v_1$ for $v \in \mathbb{R}^3$.

Lemma 2.1 (location)**.** *Every critical point of V is of the form $x = \sum w_i p_i / \sum w_i$ with $w_i = q_i|x - p_i|^{-\beta} > 0$.*

- *If the p_i are collinear, V has exactly two critical points.*
- *Otherwise, the critical points of V are exactly the critical points of $V|_\Pi$ that lie in the open triangle T.*

Proof. We have $\nabla V(x) = -2\alpha \sum_i q_i r_i^{-\beta}(x - p_i)$, with $r_i = |x - p_i|$. This gives the first claim.

Collinear case. Every critical point lies on the open segment spanned by the charges, and not at a charge. On each of the two open intervals between consecutive charges, V is strictly convex (each $|t - s|^{-2\alpha}$ is) and tends to $+\infty$ at both ends. So each interval contains exactly one critical point.

Non-collinear case. The point x is a combination of the p_i with positive weights, so it lies in T. For $x \in \Pi$ the gradient lies in Π, so critical points of V in Π are the same as critical points of $V|_\Pi$. □

From now on the charges are not collinear, and we work in $\Pi = \mathbb{R}^2$. For $x \in T$ let:

- $\lambda_i = \lambda_i(x)$ be the barycentric coordinates, $r_i = |x - p_i|$, and $u_i = (x - p_i)/r_i$;
- $\theta_i = \angle p_j x p_k$ be the angle at x subtending the side opposite p_i, where $\{i, j, k\} = \{1, 2, 3\}$;
- Δ be the area of T, A_i its angle at p_i, $a_i = |p_j p_k|$, and R, O its circumradius and circumcentre.

Then $\theta_1 + \theta_2 + \theta_3 = 2\pi$ and $\theta_i < \pi$. Since $\Delta\lambda_i$ is the area of $x p_j p_k$,

$$\lambda_i = \frac{r_j r_k \sin\theta_i}{2\Delta}. \tag{1}$$

Put

$$S(x) = \sum_i \lambda_i u_i u_i^{\mathsf{T}} \quad (\operatorname{tr} S = 1).$$

By the Cauchy–Binet formula,

$$D(x) := \det S(x) = \sum_{i<j} \lambda_i\lambda_j \,(u_i \times u_j)^2 = \sum_{\{i,j,k\}} \lambda_i\lambda_j \sin^2\theta_k. \tag{2}$$

Lemma 2.2 (reduction)**.** *Let $x \in T$. Then x is a critical point of $V = V_q$ if and only if q is proportional to $Q(x) := (\lambda_i r_i^{\beta})_{i=1}^3$. In that case x is*

- *a non-degenerate local minimum of $V|_\Pi$ if $D(x) > c$;*
- *a non-degenerate saddle if $D(x) < c$;*
- *degenerate if $D(x) = c$.*

$V|_\Pi$ has no local maxima.

Proof. By Lemma 2.1 and uniqueness of barycentric coordinates, x is critical if and only if $(q_i r_i^{-\beta})_i$ is proportional to $(\lambda_i)_i$.

In that case write $q_i r_i^{-\beta} = \kappa\lambda_i$ with $\kappa > 0$. Then

$$\operatorname{Hess} V(x) = -2\alpha \sum_i q_i r_i^{-\beta}\big(I - \beta u_i u_i^{\mathsf{T}}\big) = 2\alpha\kappa(\beta S - I).$$

If $s \le 1/2$ is the smaller eigenvalue of S, the eigenvalues of $\beta S - I$ are $\beta s - 1$ and $\beta(1-s) - 1 > 0$, using $\beta > 2$. The function $t \mapsto t(1-t)$ is increasing on $[0, 1/2]$. Hence $\beta s - 1 > 0$ holds if and only if

$$D = s(1-s) > \tfrac{1}{\beta}\big(1 - \tfrac{1}{\beta}\big) = c.$$

Likewise $\beta s - 1 < 0$ if and only if $D < c$, and $\beta s - 1 = 0$ if and only if $D = c$. □

Lemma 2.3 (index count)**.** *If all critical points of $V|_\Pi$ are non-degenerate, then*

$$\#\{\textit{saddles}\} = \#\{\textit{minima}\} + 2.$$

Proof. Let Ω be a large disc minus small discs around the p_i.

- On the outer circle, $\nabla V = -2\alpha(\sum q_i)|x|^{-\beta} x\,(1 + o(1))$ has degree 1.
- On the circle around p_i, $\nabla V = -2\alpha q_i r_i^{-\beta}(x - p_i)(1 + o(1))$ also has degree 1.

By the Poincaré–Hopf theorem, the indices of the zeros of ∇V in Ω sum to $1 - 3 = -2$. Minima have index $+1$, saddles -1, and there are no maxima by Lemma 2.2. □

3. Angle coordinates and separation of variables

Lemma 3.1. *The map $x \mapsto \theta(x) = (\theta_1, \theta_2, \theta_3)$ is injective on T.*

Proof. By the inscribed angle theorem, the points $y \in T$ with $\angle p_2 y p_3 = \theta_1$ lie on a circle Γ_1 through p_2 and p_3. Similarly, those with $\angle p_3 y p_1 = \theta_2$ lie on a circle Γ_2 through p_3 and p_1.

If $\Gamma_1 = \Gamma_2$, this circle passes through p_1, p_2, p_3, so it is the circumcircle, which does not meet the open triangle. Hence $\Gamma_1 \neq \Gamma_2$. Two distinct circles meet in at most two points, and one of them is $p_3 \notin T$. □

Lemma 3.2 (pedal identity)**.** *Let $x \in T$, put $\delta_i = \theta_i - A_i$, and let $P_x = R^2 - |x - O|^2 > 0$. Then*

(a) *$\delta_i > 0$ and $\delta_1 + \delta_2 + \delta_3 = \pi$;*
(b) *$r_j r_k \sin\delta_i = P_x \sin A_i$ for $\{i, j, k\} = \{1, 2, 3\}$.*

The δ_i are the angles of the pedal triangle of x. Part (b) is equivalent to Euler's formula $\Delta\, P_x/(4R^2)$ for the area of the pedal triangle [6]. For completeness we give a short proof.

Proof. Let ι be the inversion in the unit circle centred at x, and put $p_i^* = \iota(p_i)$. We use three standard properties of ι:

(i) $|\iota(y) - \iota(z)| = |y - z|/(|y - x|\,|z - x|)$;
(ii) the triangles xyz and $x\,\iota(z)\,\iota(y)$ are similar;
(iii) a circle with centre O and radius R not passing through x is mapped to a circle of radius $R/|R^2 - |x - O|^2|$.

Since ι preserves the rays from x, we have $\angle p_j^* x p_k^* = \theta_i$. These three angles are less than π and sum to 2π, so x lies inside $T^* = p_1^* p_2^* p_3^*$. By (ii), the angle of T^* at p_i^* is

$$\angle p_j^* p_i^* x + \angle x p_i^* p_k^* = \angle x p_j p_i + \angle x p_k p_i = (A_j - \angle x p_j p_k) + (A_k - \angle x p_k p_j) = A_j + A_k - (\pi - \theta_i) = \delta_i.$$

This proves (a).

The circumcircle of T does not pass through x, since x lies inside it. By (iii) it is mapped to the circumcircle of T^*, of radius R/P_x. The law of sines in T^* and in T, together with (i), gives

$$\frac{a_i}{r_j r_k} = |p_j^* p_k^*| = \frac{2R}{P_x}\sin\delta_i, \qquad a_i = 2R\sin A_i.$$

This is (b). □

Corollary 3.3 (separated form)**.** *For $x \in T$,*

$$\lambda_i r_i^\beta = C(x)\,\frac{\sin\theta_i\,\sin^b(\theta_i - A_i)}{\sin^b A_i}, \qquad C(x) = \frac{(r_1 r_2 r_3)^{b+1}}{2\Delta\, P_x^b} > 0. \tag{3}$$

Proof. By Lemma 3.2(b),

$$r_i = \frac{r_1 r_2 r_3}{r_j r_k} = \frac{r_1 r_2 r_3 \sin\delta_i}{P_x \sin A_i},$$

and by (1), $\lambda_i r_i = r_1 r_2 r_3 \sin\theta_i/(2\Delta)$. Now write $\lambda_i r_i^\beta = \lambda_i r_i \cdot r_i^b$. □

4. The region of minima

Let $L = \{\theta \in \mathbb{R}^3 : \theta_1 + \theta_2 + \theta_3 = 2\pi\}$ and let

$$H_A = \{\theta \in L :\ A_i < \theta_i < \pi\ (i = 1, 2, 3)\},$$

an open convex polygon. By Lemma 3.2(a), $\theta(T) \subset H_A$. For $\theta \in H_A$ define

$$W_i(\theta_i) = \frac{\sin A_i}{\sin(\theta_i - A_i)\sin\theta_i} = \cot(\theta_i - A_i) - \cot\theta_i > 0, \qquad E(\theta) = e_2(W_1, W_2, W_3).$$

Lemma 4.1. *For every $x \in T$,*

$$D(x) = \frac{P_x \sin\theta_1 \sin\theta_2 \sin\theta_3}{2\Delta} = \frac{1}{E(\theta(x))}.$$

Consequently $D(x) > c$ if and only if $E(\theta(x)) < (b+1)^2/b$.

Proof. By Lemma 3.2(b), $W_i = r_j r_k/(P_x \sin\theta_i)$. Hence $W_i W_j = r_1 r_2 r_3\, r_k/(P_x^2 \sin\theta_i \sin\theta_j)$ and

$$E = \frac{r_1 r_2 r_3}{P_x^2 \prod_l \sin\theta_l} \sum_k r_k \sin\theta_k.$$

By (1) and (2),

$$D = \frac{r_1 r_2 r_3 \prod_l \sin\theta_l}{(2\Delta)^2} \sum_k r_k \sin\theta_k.$$

Therefore

$$DE = \Big(\frac{r_1 r_2 r_3 \sum_k r_k \sin\theta_k}{2\Delta\, P_x}\Big)^2.$$

Since $r_i r_j \sin\theta_k = 2\Delta\lambda_k$, we get $r_1 r_2 r_3 \sum_k r_k \sin\theta_k = 2\Delta \sum_k \lambda_k r_k^2$. Finally

$$\sum_k \lambda_k |x - p_k|^2 = |x - O|^2 - 2\Big\langle x - O, \sum_k \lambda_k (p_k - O)\Big\rangle + R^2 = R^2 - |x - O|^2 = P_x.$$

So $r_1 r_2 r_3 \sum_k r_k \sin\theta_k = 2\Delta P_x$. Substituting this into the expressions for D and E gives $D = P_x \prod_l \sin\theta_l/(2\Delta)$ and $DE = 1$. □

Lemma 4.2 (convexity)**.** *E is strictly convex on H_A, i.e. along every non-degenerate segment in H_A. Consequently:*

- *$\widehat{M} = \{\theta \in H_A : E(\theta) < (b+1)^2/b\}$ is convex;*

- *if $E(\theta), E(\theta') \le (b+1)^2/b$, then the open segment between θ and θ' lies in $\widehat{M}$.*

Proof. Put $\ell_i(t) = \log W_i(t) = \log\sin A_i - \log\sin(t - A_i) - \log\sin t$. Then

$$\ell_i''(t) = \csc^2(t - A_i) + \csc^2 t > 0.$$

For $i \neq j$, $W_iW_j = \exp(\ell_i(\theta_i) + \ell_j(\theta_j))$ is convex, and its Hessian is at least $W_iW_j \operatorname{Hess}(\ell_i + \ell_j)$.

A nonzero direction v with $\sum v_i = 0$ has two nonzero entries, say v_i and v_j. Then $\ell_i'' v_i^2 + \ell_j'' v_j^2 > 0$, so E is strictly convex along v. The two consequences follow. □

Lemma 4.3 (definiteness)**.** *For $\theta \in H_A$ let $d_i = \cot\theta_i + b\cot(\theta_i - A_i)$. Then*

$$e_2(d) = (b+1)^2 - b\,E(\theta).$$

If $\theta \in \widehat{M}$, then $\sum_i d_i v_i^2 > 0$ for all nonzero $v \in \mathbb{R}^3$ with $\sum v_i = 0$.

Proof. If three angles in $(0, \pi)$ sum to π or to 2π, their cotangents C satisfy $e_2(C) = 1$. Indeed, $\cot(\theta_1 + \theta_2) = -\cot\theta_3$ unwinds to this.

Let $X = (\cot\delta_i)_i$ and $C = (\cot\theta_i)_i$, so $e_2(X) = e_2(C) = 1$ by Lemma 3.2(a). Put $B = \sum_{i\neq j} X_iC_j$. Since $W = X - C$ and $d = C + bX$,

$$e_2(W) = 2 - B, \qquad e_2(d) = 1 + b^2 + bB = (b+1)^2 - b\,e_2(W).$$

In the basis $(e_1 - e_3, e_2 - e_3)$ of $\{\sum v_i = 0\}$, the form $\sum d_i v_i^2$ has Gram matrix

$$G = \begin{pmatrix} d_1 + d_3 & d_3 \\ d_3 & d_2 + d_3 \end{pmatrix}, \qquad \det G = e_2(d) > 0 \ \text{ on } \widehat{M}.$$

So G is definite on $\widehat{M}$. Since $\widehat{M}$ is connected (Lemma 4.2), the eigenvalues of G never vanish there and have constant signs.

At $\theta^\circ = ((\pi + A_i)/2)_i$, which is the θ-value of the incentre, we have $W_i = 2\tan(A_i/2)$. Hence

$$E(\theta^\circ) = 4\sum_{i<j} \tan\tfrac{A_i}{2}\tan\tfrac{A_j}{2} = 4 < \frac{(b+1)^2}{b},$$

so $\theta^\circ \in \widehat{M}$. Moreover $d_i = (b-1)\tan(A_i/2) > 0$ there, so G is positive definite on $\widehat{M}$. □

5. Proof of Theorem 1.1

The collinear case is Lemma 2.1. Let the charges be non-collinear and fix q. On H_A put

$$f_i(t) = \log\sin t + b\log\sin(t - A_i), \qquad f(\theta) = (f_1(\theta_1), f_2(\theta_2), f_3(\theta_3)),$$

and let $\psi_q = (\log q_i + b\log\sin A_i)_i$. Note that $f_i' = d_i$.

Step 1: dictionary. Let $x \in T$ be a critical point of V_q and $\theta = \theta(x)$.

- By Lemma 2.2 and (3), $f(\theta) - \psi_q \in \mathbb{R}(1,1,1)$.
- By Lemmas 2.2 and 4.1, x is a minimum of $V_q|_\Pi$ or degenerate if and only if $D(x) \ge c$, if and only if $E(\theta) \le (b+1)^2/b$.

Step 2: at most one critical point with $D \ge c$. Suppose $x \neq x'$ are two such points, with $\theta = \theta(x)$ and $\theta' = \theta(x')$.

- By Lemma 3.1, $v = \theta - \theta' \neq 0$, and $\sum v_i = 0$.
- By Step 1, $f(\theta) - f(\theta') = \mu(1,1,1)$ for some $\mu \in \mathbb{R}$.

Hence

$$0 = \mu \sum_i v_i = \sum_i \big(f_i(\theta_i) - f_i(\theta_i')\big) v_i = \int_0^1 \sum_i d_i(\theta_i' + tv_i)\, v_i^2 \, dt.$$

For $0 < t < 1$ the point $\theta' + tv$ lies in $\widehat{M}$ (Lemma 4.2), so the integrand is positive (Lemma 4.3). This is a contradiction.

Step 3: at most three saddles. Let $\Psi = (\log(Q_1/Q_3), \log(Q_2/Q_3)) \colon T \to \mathbb{R}^2$.

(a) Degeneracy is singularity of $D\Psi$. Let x_0 be a critical point of V_q. Put $\varepsilon_3 = 0$ and, for $i = 1, 2$,

$$\varepsilon_i(x) = \log(q_i/q_3) - \log(Q_i(x)/Q_3(x)).$$

Then $q_i = \kappa(x) Q_i(x) e^{\varepsilon_i(x)}$ with $\kappa = q_3/Q_3 > 0$. Using $Q_i r_i^{-\beta} = \lambda_i$ and $\sum \lambda_i (x - p_i) = 0$,

$$\nabla V_q(x) = -2\alpha\kappa(x) \sum_i \lambda_i(x) \big(e^{\varepsilon_i(x)} - 1\big)(x - p_i).$$

At x_0 we have $\varepsilon = 0$, so

$$\operatorname{Hess} V_q(x_0) = 2\alpha\kappa\, \mathcal{L} \circ D\Psi(x_0), \qquad \mathcal{L}(e_1, e_2) = \lambda_1 e_1 (x_0 - p_1) + \lambda_2 e_2 (x_0 - p_2).$$

The map $\mathcal{L}$ is invertible because x_0, p_1, p_2 are not collinear. So x_0 is degenerate if and only if $D\Psi(x_0)$ is singular.

(b) Density of Morse charges. By Sard's theorem the regular values of Ψ are dense in $\mathbb{R}^2$. Charges whose log-ratios form a regular value give a Morse function $V|_\Pi$.

(c) Persistence of saddles. Suppose V_q had four distinct saddles $x_1, \dots, x_4$. They are non-degenerate and satisfy $D < c$. By the implicit function theorem applied to Ψ at the x_j, every charge vector q' close enough to q has critical points $x_1', \dots, x_4'$ near $x_1, \dots, x_4$. These are pairwise distinct and satisfy $D(x_j') < c$, so they are four saddles of $V_{q'}$.

Choose such a q' for which $V_{q'}|_\Pi$ is Morse. By Step 2, $V_{q'}$ has at most one minimum, so by Lemma 2.3 it has at most three saddles. This is a contradiction, so V_q has at most three saddles.

Conclusion. By Lemma 2.2, every critical point of $V_q|_\Pi$ either has $D \geq c$ or is a saddle. By Steps 2 and 3 there are at most $1 + 3 = 4$ critical points. By Lemma 2.1 these are all the critical points of V in $\mathbb{R}^n \setminus \{p_1, p_2, p_3\}$. □

6. Remarks on Part I

Remark 6.1 (a hyperbolic reading). The cotangent vectors of the angles of a triangle lie on the sheet $\{e_2 = 1,\ \sum > 0\}$. Equip $\mathbb{R}^3$ with the polarisation $\langle u, v\rangle = \frac{1}{2}\sum_{i \neq j} u_i v_j$ of e_2. Then this sheet is a hyperboloid model of the hyperbolic plane. Indeed, the triangle with vertices $0, 1, z$ ($\operatorname{Im} z > 0$) has cotangent vector

$$\big(\operatorname{Re} z,\ 1 - \operatorname{Re} z,\ |z|^2 - \operatorname{Re} z\big) / \operatorname{Im} z,$$

and this map is an isometry from the Poincaré half-plane.

Take $X = (\cot \delta_i)$, the shape of the pedal triangle, and $Y = (\cot(\pi - \theta_i))$, the shape of the triangle with angles $\pi - \theta_i$. The proof of Lemma 4.3 shows that $E = 2 + 2\langle X, Y\rangle = 2 + 2\cosh\rho$, where ρ is the hyperbolic distance between these two shapes. Thus

$$D = \frac{1}{4\cosh^2(\rho/2)},$$

and x is a minimum for its charges if and only if $\rho < \log(2\alpha + 1)$. The classical quantity $|Z|$, $Z = \sum_i \lambda_i e^{2i \arg u_i}$, $D = (1 - |Z|^2)/4$, equals $\tanh(\rho/2)$.

Remark 6.2. (i) *The angle map is a bijection $\theta\colon T \to H_A$, with explicit inverse*

$$\lambda_i \propto \frac{\sin A_i \sin\theta_i}{\sin(\theta_i - A_i)}.$$

Indeed, by (1) and Lemma 3.2(b), every $x \in T$ satisfies

$$\lambda_i = \frac{P_x \sin A_i \sin\theta_i}{2\Delta \sin\delta_i}.$$

Given $\theta \in H_A$, the weights $\sin A_i \sin\theta_i / \sin(\theta_i - A_i)$ are positive, so they define a point $x \in T$. Then $\theta(x) = \theta$, because $\theta \mapsto [\sin\theta_i/\sin(\theta_i - A_i)]_i$ is injective on H_A: each coordinate is decreasing in θ_i, and $\sum \theta_i$ is fixed.

(ii) *The incentre is the unique maximum point of D, where $D = 1/4$.* Each W_i has a critical point at $\theta_i^\circ = (\pi + A_i)/2$, so θ° is a critical point of the strictly convex function E on H_A. Hence it is the unique minimum point of E, with $E(\theta^\circ) = 4$.

(iii) *V_q has a non-degenerate planar minimum if and only if $\psi_q \in f(\widehat{M}) + \mathbb{R}(1,1,1)$.* This follows from Step 1 and (i). In that case all other critical points have $D < c$ by Step 2. So $V_q|_\Pi$ is Morse, and by Lemma 2.3 it has exactly four critical points. Moreover f is injective on $\widehat{M}$ by Step 2.

Remark 6.3. We checked all identities used in Part I numerically at random triangles and points, with relative errors below 10^{-9}. We also compared the predictions of Theorem 1.1 and Lemma 4.1 with direct critical-point searches, over several thousand configurations and several values of α, and found no discrepancy.

Part II. Charges of different signs

7. Setting

In Part II, write $\alpha = \frac{1}{2}$ (or $\alpha \in [\frac{1}{4}, 2]$ for Theorem 1.4), $b = 2\alpha + 1$, $\beta = b + 1$, $c = b/(b+1)^2$ and $K = (b+1)^2/b$. Put $F = K/2 - 1$; for Coulomb, $K = \frac{9}{2}$ and $F = \frac{5}{4}$.

Replacing V by $-V$ and relabelling, we may assume $q_1, q_2 > 0 > q_3$. The triangle $p_1p_2p_3$ is labelled counter-clockwise. It has angles A_i, area Δ, circumradius R and circumcentre O. Put

$$P_x = R^2 - |x - O|^2.$$

For x off the side lines, let λ_i be the signed barycentric coordinates and $r_i = |x - p_i|$. Let $\sigma_i \in (-\pi, \pi)$ be the signed angle at x from $p_j - x$ to $p_k - x$, for (i,j,k) cyclic, so that $\lambda_i = r_j r_k \sin\sigma_i/(2\Delta)$. Write $e_2(v) = v_1v_2 + v_2v_3 + v_3v_1$.

Lemma 7.1 (planarity and type)**.** *Let the charges be non-collinear.*

1. *Every critical point of V lies in the plane Π of the charges.*
2. *A point $x \in \Pi$ is critical if and only if $(q_i r_i^{-\beta})_i = \kappa(\lambda_i)_i$ for some $\kappa \neq 0$.*
3. *Write $S = \sum_i \lambda_i u_i u_i^{\mathsf{T}}$ (so $\operatorname{tr} S = 1$) with $u_i = (x - p_i)/r_i$, and $D = \det S$. Then $\operatorname{Hess} V|_\Pi(x) = 2\alpha\kappa(\beta S - I)$. Hence x is an extremum of $V|_\Pi$ if and only if $D > c$: a minimum if $\kappa > 0$, a maximum if $\kappa < 0$. If $D < c$, then x is a non-degenerate saddle.*

Proof. (1) Write $x = y + hn$ with $y \in \Pi$ and $n \perp \Pi$, and put $w_i = q_i r_i^{-\beta} \neq 0$. Then $\nabla V = -2\alpha\big(\sum w_i(y - p_i) + hn \sum w_i\big)$. If $h \neq 0$, then $\sum w_i = 0$ and $\sum w_i p_i = 0$, which forces $w = 0$ because the p_i are affinely independent; this is impossible.

(2) This is the same computation, by uniqueness of barycentric coordinates.

(3) $\operatorname{Hess} V = -2\alpha \sum w_i (I - \beta u_i u_i^{\mathsf{T}})$. The eigenvalues of S are s and $1-s$; if $s \le \frac{1}{2}$, then $\beta(1-s) - 1 > 0$ because $\beta > 2$. So the Hessian is definite if and only if $s > 1/\beta$, which is equivalent to $D = s(1-s) > c$. □

Lemma 7.2 (signed identities)**.** *For every x off the side lines:*

(P) $r_j r_k \sin(\sigma_i - A_i) = P_x \sin A_i$*;*
(D) $D(x) = P_x \sin\sigma_1 \sin\sigma_2 \sin\sigma_3/(2\Delta)$*;*
(E) $D(x)\, e_2(W) = 1$*, where* $W_i = \cot(\sigma_i - A_i) - \cot\sigma_i$*.*

Proof. With $p_1 = (0,0)$, $p_2 = (1,0)$ and $p_3 = (u,v)$, each identity becomes a polynomial identity in x, u, v and the square roots r_i and $|p_j - p_k|$, once these roots are treated as symbols subject to their squares.

- For (P), multiply by $|p_i - p_j||p_i - p_k|$.
- For (D), multiply by $(r_1 r_2 r_3)^2$.
- For (E), clear denominators.

Expanding, each difference vanishes identically (`code/symbolic_mixed.py`). For x inside the triangle these are Lemma 3.2(b) and Lemma 4.1. □

Corollary 7.3. *Let $R_+ = \{\lambda_1, \lambda_2 > 0 > \lambda_3\}$ (beyond the side p_1p_2, inside the angle at p_3) and $R_- = \{\lambda_1, \lambda_2 < 0 < \lambda_3\}$ (the vertical angle at p_3). Every critical point lies in $R_+ \cup R_-$, and those in R_- are non-degenerate saddles. Minima, and degenerate critical points, lie in*

$$R_+^{\text{out}} = R_+ \cap \{P_x < 0\}.$$

Proof. By Lemma 7.1(2), the signs of $\lambda(x)$ are $\pm(+,+,-)$.

In R_-, $\operatorname{sign}\sin\sigma_i = \operatorname{sign}\lambda_i$, so $\prod \sin\sigma_i > 0$. Moreover $P_x < 0$: p_3 is a convex combination of x, p_1, p_2 with positive weights, so if x were in the closed circumdisc, p_3 would lie in the open disc. By (D), $D < 0 < c$.

In R_+, $\prod \sin\sigma_i < 0$, so $D \ge c > 0$ forces $P_x < 0$. □

8. Angle coordinates and the gradient structure

For $x \in R_+$ the segment p_3x crosses p_1p_2. So $\sigma_1 = t_1 \in (0,\pi)$, $\sigma_2 = t_2 \in (0,\pi)$ and $\sigma_3 = -s$, where $s = t_1 + t_2$. On R_+^{out}, (P) gives $\sin(t_i - A_i) < 0$, so

$$t \in Q := (0, A_1) \times (0, A_2).$$

Lemma 8.1. *The map $x \mapsto t = (t_1, t_2)$ is injective on R_+^{out}.*

Proof. The point x is the second intersection, other than p_3, of the circle through p_2, p_3 determined by t_1 and the circle through p_3, p_1 determined by t_2. These circles are distinct: otherwise their common circle would be the circumcircle, which does not contain x. □

By (P), $r_i = r_1 r_2 r_3 |\sin(\sigma_i - A_i)|/(|P_x| \sin A_i)$. Hence

$$\lambda_i r_i^{\beta} = C(x)\, \sin\sigma_i\, |\sin(\sigma_i - A_i)|^b / \sin^b A_i, \qquad C(x) > 0.$$

Put

$$\varphi_i(t) = \log\sin t + b \log\sin(A_i - t)\ (i = 1,2), \qquad \varphi_3(s) = \log\sin s + b\log\sin(s + A_3),$$

and

$$H(t) = \Phi_1(t_1) + \Phi_2(t_2) - \Phi_3(t_1 + t_2), \qquad \Phi_i' = \varphi_i.$$

Then $x \in R_+^{\text{out}}$ is critical for q if and only if $\nabla H(t(x)) = \psi_q$, where

$$\psi_q = \big(\log\tfrac{q_1}{|q_3|} + b\log\tfrac{\sin A_1}{\sin A_3},\ \log\tfrac{q_2}{|q_3|} + b\log\tfrac{\sin A_2}{\sin A_3}\big).$$

9. The region of minima outside the triangle

Let $U_i(t) = \cot t + \cot(a_i - t)$ with $a_1 = A_1$, $a_2 = A_2$, $a_3 = A_1 + A_2$, and evaluate U_3 at s. In the coordinates t we have $W = (-U_1, -U_2, U_3)$, so by (E)

$$E := U_1U_2 - U_3(U_1 + U_2) = 1/D. \tag{4}$$

Thus minima and degenerate points correspond to points of $\overline{M'}$, where $M' = \{t \in Q : E(t) < K\}$.

Lemma 9.1. (1) *$E \geq 4$, with equality exactly at $t^\circ = (A_1/2, A_2/2)$, which corresponds to the excentre opposite p_3.*

(2) *With $d = (\varphi_1', \varphi_2', -\varphi_3')$ we have $e_2(d) = (b+1)^2 - bE$. So* Hess H *is definite on M'; if M' is connected, it is negative definite there.*

Proof. Put $\tilde{x} = (\cot t_1, \cot t_2, -\cot s)$ and $\tilde{y} = (\cot(A_1 - t_1), \cot(A_2 - t_2), -\cot(A_1 + A_2 - s))$. Both satisfy $e_2 = 1$, by the cotangent addition formula. Also $\tilde{x} + \tilde{y} = (U_1, U_2, -U_3)$ and $d = \tilde{x} - b\tilde{y}$.

Polarising e_2 gives $E = 2 + 2\langle \tilde{x}, \tilde{y} \rangle$ and $e_2(d) = 1 + b^2 - b(E-2)$, where $\langle \cdot, \cdot \rangle$ is the Lorentzian polarisation of e_2 (Remark 6.1). Moreover $\tilde{x}$ and $\tilde{y}$ are the cotangent vectors of the genuine triangles with angles $(t_1, t_2, \pi - s)$ and $(A_1 - t_1, A_2 - t_2, A_3 + s)$. So both lie on the sheet $\{e_2 = 1, \sum > 0\}$ of the hyperboloid, hence $\langle \tilde{x}, \tilde{y} \rangle \geq 1$, with equality iff $\tilde{x} = \tilde{y}$, i.e. iff $t = t^\circ$. (Only $E(t^\circ) = 4 < K$ is used below.)

At t° we have $d = \big((1-b)\cot\frac{A_1}{2}, (1-b)\cot\frac{A_2}{2}, (b-1)\cot\frac{A_1+A_2}{2}\big)$, which has two negative entries. With $e_2(d) > 0$, this means Hess $H = \operatorname{diag}(d)|_{\sum v = 0}$ is negative definite. □

Lemma 9.2 (index count)**.** *If $V|_\Pi$ is Morse, then $\#\{\text{saddles}\} = \#\{\text{extrema}\} + 2$ when $q_1 + q_2 + q_3 \neq 0$, and $\#\{\text{extrema}\} + 1$ otherwise.*

Proof. We apply Poincaré–Hopf to ∇V on a large disc minus small discs around the charges.

- On each small circle, ∇V is radial, so it has degree 1 whatever the sign of the charge.
- On a large circle the degree is 1 when the total charge is nonzero.
- When the total charge is zero, the dipole moment $m = \sum q_i p_i$ is nonzero, and the field is asymptotic to $\nabla(\langle m, x \rangle |x|^{-\gamma})$, $\gamma = 2\alpha + 2 > 2$. On circles this field is proportional to $(1 - \frac{\gamma}{2}) - \frac{\gamma}{2} e^{2i\phi}$, which has degree 2.

Extrema have index $+1$ and saddles index -1. □

10. An explicit description of the region

Put $\Lambda_i(t) = \log\big(\sin t / \sin(a_i - t)\big)$. Then $\Lambda_i' = U_i$, $U_i = 2(\cosh\Lambda_i + \cos a_i)/\sin a_i$, and $t = \arg(1 + e^{\Lambda + ia})$.

Lemma 10.1 (product representation)**.** *With $\Delta := \Lambda_1(t_1) - \Lambda_2(t_2)$ (from here on, Δ denotes this difference, not the area),*

$$E = \frac{2U_3(s)}{\sin A_3}(\cosh\Delta + \cos A_3) = \frac{2(\cosh\Delta + \cos A_3)}{\sin s \, \sin(s + A_3)}.$$

Similarly

$$E = \frac{2\big(\cosh(\Lambda_2(t_2) - \Lambda_3(s)) - \cos A_1\big)}{\sin t_1 \, \sin(A_1 - t_1)}, \qquad \Lambda_3(s) = \log\frac{\sin s}{\sin(A_1 + A_2 - s)}.$$

Proof. Geometrically, $E = 2 + 2\cosh\rho$, where ρ is the hyperbolic distance between the shapes $\tilde{x}, \tilde{y}$. The formula is the distance formula of the strip model of the hyperbolic plane, anchored at vertex 3 (resp. 1).

For a direct proof, write every sine and cosine through e^{it_1}, e^{it_2}, e^{iA_1}, e^{iA_2}. Both identities then become identities between rational functions, and they are verified exactly in `code/symbolic_product.py`. □

Since $1/(\sin s \sin(s + A_3)) = U_3/\sin A_3$, Lemma 10.1 gives

$$E < K \iff \cosh\Delta < \beta(s) := \tfrac{1}{2}\big[(F-1)\cos A_3 - (F+1)\cos(2s + A_3)\big]. \tag{5}$$

Lemma 10.2. (1) *The map $t \mapsto (s, \Delta)$ is a diffeomorphism of Q onto $(0, A_1 + A_2) \times \mathbb{R}$.*

(2) *Write $2s + A_3 = \pi + \psi$. Then $\beta \geq 1$ exactly when $|\psi| \leq \psi_{\max}$, where $\cos\psi_{\max} = (2 - (F-1)\cos A_3)/(F+1)$, and $\psi_{\max} < A_1 + A_2$.*

(3) *Consequently $M' = \{|\Delta| < \operatorname{arccosh}\beta(s)\}$ is a topological disc, and $\partial M' = \{\cosh\Delta = \beta(s)\}$ is a smooth simple closed curve.*

Proof. (1) For fixed s, Δ increases strictly with t_1 along $t_1 + t_2 = s$, from $-\infty$ to $+\infty$. Its derivative is $U_1 + U_2 > 0$.

(2) The bound $\psi_{\max} < A_1 + A_2 = \pi - A_3$ is equivalent to $\cos A_3 > -1$.

(3) The gradient $(-\beta', \sinh\Delta)$ does not vanish on the curve: if $\Delta = 0$, then $\beta = 1 < \max\beta$, so $\beta' \neq 0$. □

Given (s, Δ), the point t is explicit: $\zeta = e^{\Lambda_1}$ is the positive root of

$$e^{-\Delta}\sin(A_1 + A_2 - s)\,\zeta^2 + \big[\sin(A_1 - s) + e^{-\Delta}\sin(A_2 - s)\big]\zeta - \sin s = 0, \tag{6}$$

which expresses $\arg(1 + \zeta e^{iA_1}) + \arg(1 + \zeta e^{-\Delta} e^{iA_2}) = s$.

11. The curvature criterion

Lemma 11.1. *In linear coordinates X, let $\hat{G}(X) = \beta(X_1 + X_2) - \cosh\big(\tilde{\Lambda}_1(X_1) - \tilde{\Lambda}_2(X_2)\big)$. Write $\tilde{U}_i = \tilde{\Lambda}_i'$ (of any sign), $S = \sinh\Delta$, $C = \cosh\Delta$ and $P = -\beta''$. On $\{\hat{G} = 0\}$, the vector $\tau = (\beta' + S\tilde{U}_2,\ S\tilde{U}_1 - \beta') \neq 0$ is tangent, and*

$$-\operatorname{Hess}\hat{G}(\tau, \tau) = J := (\tilde{U}_1 + \tilde{U}_2)^2(PS^2 + C\beta'^2) + S\big[\tilde{U}_1'(\beta' + S\tilde{U}_2)^2 - \tilde{U}_2'(S\tilde{U}_1 - \beta')^2\big].$$

Hence the boundary of $\{\hat{G} > 0\}$ has positive curvature at a point if and only if $J > 0$ there. In particular, $\{\hat{G} > 0\}$ is strictly convex near such a point.

Proof. We have $\nabla\Delta = (\tilde{U}_1, -\tilde{U}_2)$ and $\operatorname{Hess}\Delta = \operatorname{diag}(\tilde{U}_1', -\tilde{U}_2')$, so

$$\operatorname{Hess}\hat{G} = \beta''\mathbf{1}\mathbf{1}^{\mathsf{T}} - C\nabla\Delta\nabla\Delta^{\mathsf{T}} - S\operatorname{Hess}\Delta.$$

The vector τ is orthogonal to $\nabla\hat{G} = \beta'\mathbf{1} - S\nabla\Delta$. Moreover $\tau_1 + \tau_2 = S(\tilde{U}_1 + \tilde{U}_2)$ and $\tilde{U}_1\tau_1 - \tilde{U}_2\tau_2 = \beta'(\tilde{U}_1 + \tilde{U}_2)$. Substituting gives J. □

We use Lemma 11.1 in two coordinate systems.

Vertex 3. Take $X = t$, $\tilde{U}_i = U_i > 0$, $\tilde{\Lambda}_i = \Lambda_i$. Dividing by $(U_1U_2)^2$,

$$I = (c_1 + c_2)^2(PS^2 + C\beta'^2) + S\big[a_1(\beta' c_2 + S)^2 - a_2(S - \beta' c_1)^2\big], \tag{7}$$

where $c_i = 1/U_i = \sin A_i/(2h_i)$, $a_i = U_i'/U_i^2 = \sinh\Lambda_i/h_i$ and $h_i = \cosh\Lambda_i + \cos A_i$. The expression I is invariant under exchanging the indices $1, 2$ together with $S \mapsto -S$. So the arc $\Delta < 0$ of a triangle is the arc $\Delta > 0$ of the mirrored triangle.

Vertex 1. Take $X = (s, -t_2)$, $\tilde{\Lambda}_1 = \Lambda_3$ and $\tilde{\Lambda}_2(X_2) = \Lambda_2(-X_2)$, so that $\tilde{U}_1 = U_3$ and $\tilde{U}_2 = -U_2$. Use the representation $\cosh(\Lambda_3 - \Lambda_2) < \beta_1(t_1)$, where $\beta_1 = \frac{1}{2}[(F+1)\cos\psi_1 - (F-1)\cos A_1]$ and $\psi_1 = 2t_1 - A_1$. Dividing by U_2^2,

$$J_1/U_2^2 = (1 - U_3 c_2)^2 (PS^2 + C\beta_1'^2) + S\big[U_3'(\beta_1' c_2 - S)^2 - a_2(SU_3 - \beta_1')^2\big], \tag{8}$$

with $U_3' = -U_3(\cot s + \cot(s + A_3))$.

Remark 11.2. These formulas were also tested against the curvature of $\{E = K\}$ computed directly from E. At $\alpha = 10$, where M' is not always convex, the signs agree at every one of several thousand boundary points.

12. The computer-assisted step

Proposition 12.1. *For every $\alpha \in [\frac{1}{4}, 2]$ and every triangle, $\partial M'$ has strictly positive curvature. Hence $\overline{M'}$ is strictly convex.*

Description of the verification. Parametrise the boundary by $\rho \in [-1, 1]$, with $\psi = \rho\psi_{\max}$ and the arc sign of Δ. Then S, $C = \beta$, $\beta' = -(F+1)\sin\psi$ and $P = 2(F+1)\cos\psi$ are explicit. Moreover $\beta - 1 = (F+1)\sin\frac{\psi_{\max}(1+\rho)}{2}\sin\frac{\psi_{\max}(1-\rho)}{2}$, and ζ is given by (6). We cover the space of triangles by three charts.

Main chart. $A_1 + A_2 \in [e_0, \pi]$ and $A_1, A_2 \le \pi - r_0$; we verify $I > 0$ on the arc $\Delta > 0$. The other arc follows by the mirror symmetry.

Band chart. $A_1 + A_2 = e \in [0, e_0]$, the small triangles. Here $I = O(e^4)$, and we verify $I/e^4 > 0$ in scaled variables that are regular at $e = 0$:

- $\psi_{\max} = 2\arcsin\big(q\sin\frac{e}{2}\big)$ with $q = \sqrt{(F-1)/(F+1)}$;
- S, β', c_i and Δ are each e times an analytic function, written with $\frac{\sin x}{x}$, $\frac{\arcsin x}{x}$ and $\frac{\operatorname{arcsinh} x}{x}$;
- the difference $(a_1 - a_2)/e$ is enclosed by the mean value theorem.

Corner chart. $A_1 = \pi - r$ with $r \le r_0$, $A_2 = \theta r$ and $A_3 = (1-\theta)r$. In the vertex-3 form some quantities blow up here, so we verify (8) on both arcs, where every quantity stays bounded. The boundary point is explicit:

- $\zeta = e^{\Lambda_2}$ is the positive root of $e^{\Delta}\sin(A_1 - t_1)\zeta^2 + [e^{\Delta}\sin(A_1 + A_2 - t_1) - \sin(A_2 + t_1)]\zeta - \sin t_1 = 0$;
- $t_2 = \arg(1 + \zeta e^{iA_2})$, which equals $\arctan\big(\zeta\sin A_2/(1 + \zeta\cos A_2)\big)$ here, since $A_2 \le r_0 < \pi/2$.

The corner $A_2 \ge \pi - r_0$ is the mirror image of this chart.

The computation uses interval arithmetic in IEEE binary64:

- every operation is rounded outward by two units in the last place, and every library transcendental by four, assuming the math library is accurate to two units;
- each box of parameters receives an enclosure of the quantity to be verified, and boxes whose lower bound is not positive are bisected.

The main chart takes $A_1 + A_2$ up to the double just above π, so that triangles with arbitrarily small $A_3 > 0$ are covered.

For Coulomb, with $(e_0, r_0) = (0.2, 0.05)$, all boxes are certified: 127,403 (main), 2,048 (band) and $2 \times 1{,}024$ (corner), in a few seconds. The same holds for $(e_0, r_0) = (0.1, 0.1)$ and $(0.4, 0.02)$.

For Coulomb, every certified box was also re-verified with the multiple-precision interval arithmetic of `mpmath.iv` (80-bit precision), using an independent implementation of the formulas. That run does not rely on the accuracy assumption for the math library.

Other exponents. The verified quantities depend on α only through F. Since $F = (b+1)^2/(2b) - 1$ increases from $F(\frac{1}{4}) = \frac{13}{12}$ to $F(2) = \frac{13}{5}$, we add F as a fourth interval variable. The steps are as follows.

- The interval $[\frac{1}{4}, 2]$ is cut into 163 slices of α, of widths from 0.05 down to 2^{-9}, narrower for larger α.
- On each slice, the three charts are verified with $(e_0, r_0) = (0.4, 0.05)$, for all F in a floating-point box around the values of F on the slice. In total there are about 1.3×10^9 boxes, and the computation takes about four hours on one core.
- A separate run covers the strip of the main chart with $A_1 + A_2$ between $\pi - 10^{-6}$ and the double just above π, for all $F \in [\frac{13}{12}, \frac{13}{5}]$.
- Finally, we check that the certified F-boxes cover $[\frac{13}{12}, \frac{13}{5}]$. The two ends are compared with these rationals exactly, so no error analysis of F in floating point is needed.

The enclosures with interval F were also tested against an independent evaluation at 40–60 digits, which locates the boundary point by bisection. At about 4,000 random points, every value was contained in its enclosure.

Unlike the Coulomb case, the computation for other exponents was not repeated in multiple-precision arithmetic. It relies on the accuracy assumption for the math library. The code is listed in Section 15. □

13. Proof of Theorems 1.2 and 1.4

(a) Non-collinear charges. By Lemma 7.1(1) and Corollary 7.3, every critical point lies in Π, those in R_- are saddles, and the minima and degenerate points lie in R_+^{out}.

By Lemma 8.1 and Section 8, these correspond injectively to points $t \in \overline{M'}$ with $\nabla H(t) = \psi_q$.

Suppose there were two such points, $t \neq t'$. By Proposition 12.1 the open segment between them lies in M', and by Lemma 9.1 $\operatorname{Hess} H \prec 0$ there. Hence

$$\langle \nabla H(t) - \nabla H(t'), t - t' \rangle = \int_0^1 (t - t')^{\mathsf{T}} \operatorname{Hess} H\,(t - t')\, d\lambda < 0,$$

which contradicts $\nabla H(t) = \nabla H(t')$. So there is at most one minimum or degenerate critical point.

For the saddles, Step 3 of Section 5 applies, with $\log|Q_i/Q_3|$ in place of $\log(Q_i/Q_3)$, with Ψ defined on $R_+ \cup R_-$, and with κ of either sign. Degenerate critical points are those at which $D\Psi$ is singular, so Morse charges are dense. If V had four saddles, they would persist for a nearby Morse charge vector. That contradicts Lemma 9.2 and the bound on extrema. Altogether there are at most four critical points.

(b) Collinear charges. Off the line, $\nabla V = 0$ forces $\sum w_i = 0 = \sum w_i t_i$, where t_i are the positions on the line. By rotational symmetry about the line, such equilibria form circles in $\mathbb{R}^3$, and $(n-2)$-spheres in $\mathbb{R}^n$.

A non-degenerate equilibrium on the line has Hessian $\operatorname{diag}(f'', h, h)$, so it is also non-degenerate for $V|_\Pi$, for any plane Π through the line. Moving p_3 slightly off the line inside Π, these equilibria persist as distinct critical points of a non-collinear configuration (implicit function theorem). By (a) there are at most four. □

14. Remarks on Part II

Remark 14.1 (large exponents). For large α, M' is not convex for some triangles. For example, at $\alpha = 10$ the boundary of M' has negative curvature for the triangle with angles $(10.6^\circ, 29.3^\circ, 140.1^\circ)$.

Numerically, the scale-free quantity $I/(A_1+A_2)^4$ is positive for every triangle when $\alpha \le 5.6$. It becomes negative for some nearly degenerate triangles, with one angle close to π, when $\alpha \ge 5.8$. These values were confirmed at 40–50 digits (`convexity_threshold.py`). So between $\alpha = 2$ and about 5.7, the range of Theorem 1.4 is limited by computing time rather than by a failure of convexity. Near $\alpha \approx 5.75$ the margin tends to 0.

At the other end, as $\alpha \to 0$ the level K decreases to $4 = \min E$, so M' shrinks to the point t°. The charts would have to be rescaled around t° there.

Even beyond this threshold, the charge map remains injective on M' numerically: the image of $\partial M'$ winds at most once around any point. So Maxwell's bound for three charges seems to hold for all α, but a different argument would be needed to prove it.

Remark 14.2. Numerically, three collinear charges of different signs have 0 or 2 equilibria on the line.

Part (b) uses $n \ge 3$. In $\mathbb{R}^2$ the off-axis equilibria of collinear charges are isolated mirror pairs, and they are not covered by (b).

Part (b) counts only non-degenerate equilibria, and isolated degenerate ones do occur. For charges $-2, 24, -64$ at the points $0, 1, 3$ of a line in $\mathbb{R}^3$, the point -1 of the line is a critical point with vanishing Hessian. No circle of critical points comes near it, so it is an isolated equilibrium.

15. Code

The code is available, in the folder `code`, at

`https://github.com/anshM123/Maxwell-s-Problem-on-the-Number-of-Equilibria-of-Point-Charges`

The numerical checks of Part I are `key_identities.py`, `separable.py`, `predict_test.py` and `scan.py`. The verification for Part II consists of:

- `ia.py`: interval arithmetic;
- `mixed_cert.py`: the main chart;
- `mixed_charts.py`: the band and corner charts;
- `mixed_cert_all.py`: the driver;
- `mixed_certF.py`: other exponents, with F as an interval variable;
- `mixed_alpha_sweep.py`: the sweep over α;
- `mixed_certF_sliver.py`: the strip next to $A_1 + A_2 = \pi$, for all F;
- `sweep_coverage.py`: checks that the certified F-boxes cover $[\frac{13}{12}, \frac{13}{5}]$;
- `check_certF_containment.py`: a test of the enclosures for interval F against an independent high-precision evaluation;
- `convexity_threshold.py`: the numerics of Remark 14.1;
- `symbolic_mixed.py`: the identities of Lemma 7.2;
- `symbolic_product.py`: the identities of Lemma 10.1.

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